

Ongoing Work on Contextual Neural Machine Translation

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Motivation

State of the Art Literature

Ongoing Work at i6

Experiments

Summary and Future Work

- ▶ Most neural machine translation (NMT) systems translate sentence by sentence
- ▶ Make use of other sentences in addition to exploit context information
 - ▷ reduce ambiguities in translation choices
 - ▷ example: cross-sentence anaphora [Voita & Sennrich⁺ 18]

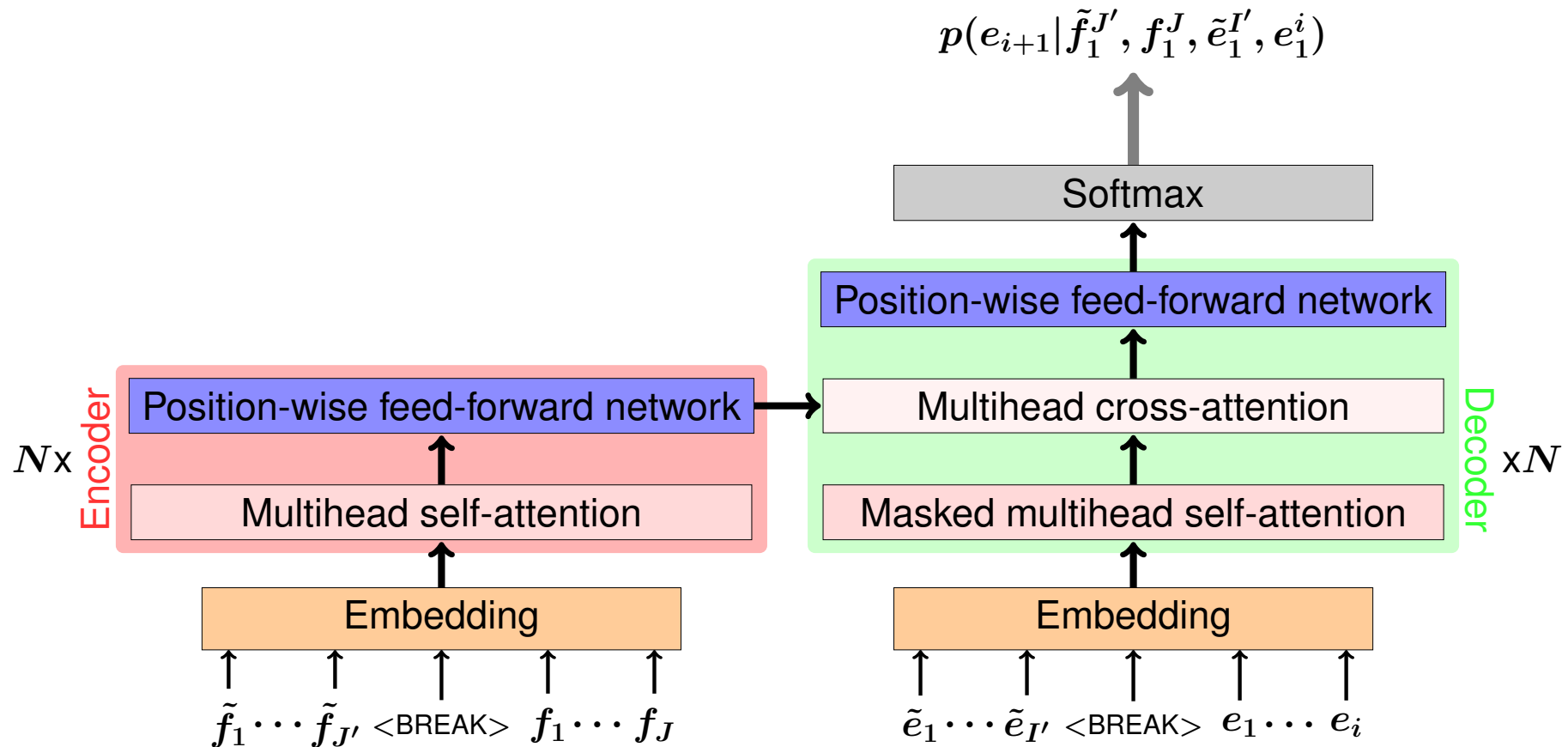
An English→German example of a cross-sentence anaphora problem. The translation of the pronoun '**it**' depends on the previous sentence.

Source sentences	Correct translation of ' it '
<i>The monkey ate the banana. Because it was hungry.</i>	er
<i>The monkey ate the banana. Because it ripe.</i>	sie
<i>The monkey ate the banana. Because it was tea-time.</i>	es

- ▶ **Single encoder: concatenate sentences on source-/target side**
[Tiedemann & Scherrer 17, Agrawal & Turchi⁺ 18]
- ▶ **Multi-Encoder with attention mechanism**
 - ▷ **Feed surrounding sentences into separate encoders and combine context vectors:**
 - **concatenation (like in [Zoph & Knight 16])**
 - **gated sum [Bawden & Sennrich⁺ 18]**
 - **hierarchical attention [Bawden & Sennrich⁺ 18]**

Remark: above methods only consider *local* context

Transformer Architecture with Concatenated Context Sentences



Transformer Architecture with concatenated sentences as additional context-information. Here we consider one previous context-sentence on the source ($\tilde{f}_1^{J'}$) and target ($\tilde{e}_1^{I'}$) side. The $<BREAK>$ token separates the sentences.

1. Reproduce results from [Agrawal & Turchi⁺ 18]

- ▶ using Transformer model as well
- ▶ using data set of IWSLT 2017 English→Italian
- ▶ concatenate sentences with special token **<BREAK>**,
example: sentence_1 **<BREAK>** sentence_2

2. compress context-sentences by pruning

- ▶ first n most-common words in training corpus
- ▶ words that occur more than n times in training corpus
- ▶ pre-defined stopwords list

3. Test method of concatenation (+ pruning) on WMT 2018 German→English data as well

Ongoing Work at i6 - Concatenation

Concatenation configurations. We denote string-concatenation by $\oplus$.

(a) Single target sentence.

Source	$\rightarrow$	Target
1 previous $\oplus$ current	$\rightarrow$	current
2 previous $\oplus$ current	$\rightarrow$	current
1 previous $\oplus$ current $\oplus$ 1 next	$\rightarrow$	current
current	$\rightarrow$	current

(b) Multiple target sentence.

Source	$\rightarrow$	Target
1 previous $\oplus$ current	$\rightarrow$	1 previous $\oplus$ current
1 previous $\oplus$ current $\oplus$ 1 next	$\rightarrow$	1 previous $\oplus$ current

Experiments - Data

Corpus statistics for both IWSLT 2017 English→Italian and WMT 2018 German→English experiments.

		IWSLT'17 EN-IT		WMT'18 DE-EN	
		EN	IT	DE	EN
Train.	Sentences	232K		5.9M	
	Running words	4.7M	4.4M	160.9M	157.7M
Dev.		<i>tst2010</i>		<i>newstest2016</i>	
	Sentences	1.5K		3K	
	Running words	31K	29K	77K	73K
Test		<i>tst2017</i>		<i>newstest2017</i>	
	Sentences	1.1K		3K	
	Running words	22K	20K	76K	73K

Experiments - Results: IWSLT 2017 English→Italian

Evaluation results on the test set *tst2017* with concatenation ($\oplus$). We report de-tokenized and cased BLEU scores.

			BLEU [%]	TER [%]
current	→	current	29.89	52.04
1 previous $\oplus$ current	→	current	30.02	51.82
2 previous $\oplus$ current	→	current	29.81	52.20
1 previous $\oplus$ current $\oplus$ 1 next	→	current	29.32	52.78
current $\oplus$ 1 next	→	current	29.93	51.87
1 previous $\oplus$ current	→ 1 previous $\oplus$ current		30.64	50.95
1 previous $\oplus$ current $\oplus$ 1 next	→ 1 previous $\oplus$ current		30.64	51.38

Evaluation results on the test set *tst2017* with concatenation ($\oplus$) and a *single* target sentence. We report de-tokenized and cased BLEU scores. We result best pruning results for each concatenation configuration.

Source	→ Target	BLEU [%]	TER [%]
current	→ current	29.89	52.04
1 previous $\oplus$ current	→ current	30.02	51.82
+ with most-common-100 pruning (59% pruned words)		30.03	52.04
2 previous $\oplus$ current	→ current	29.81	52.20
+ with most-common-90 pruning (57% pruned words)		30.35	51.38
1 previous $\oplus$ current $\oplus$ 1 next	→ current	29.32	52.78
+ with most-common-90 pruning (57% pruned words)		30.14	51.77
current $\oplus$ 1 next	→ current	29.93	51.87
+ with most-common-100 pruning (59% pruned words)		30.00	51.62

Experiments - Results: IWSLT 2017 English→Italian

Evaluation results on the test set *tst2017* with concatenation ($\oplus$) and *multiple* target sentences. We report de-tokenized and cased BLEU scores. We result best pruning results for each concatenation configuration.

Source	→	Target	BLEU [%]	TER [%]
1 previous $\oplus$ current	→	1 previous $\oplus$ current	30.64	50.95
+ with most-common-90 pruning (57% pruned words)			30.09	52.08
1 previous $\oplus$ current $\oplus$ 1 next	→	1 previous $\oplus$ current	30.64	51.38
+ with most-common-150 pruning (64% pruned words)			30.18	51.64

Experiments - Results: WMT 2018 German→English

Evaluation results on the test set *newstest2017* with concatenation. We report de-tokenized and cased BLEU scores.

			BLEU [%]	TER [%]
current	→	current	30.36	50.99
1 previous $\oplus$ current	→	current	30.13	51.44
2 previous $\oplus$ current	→	current	30.32	51.23
1 previous $\oplus$ current $\oplus$ 1 next	→	current	30.28	51.23
current $\oplus$ 1 next	→	current	30.15	51.01
1 previous $\oplus$ current	→ 1 previous $\oplus$ current		30.10	51.16
1 previous $\oplus$ current $\oplus$ 1 next	→ 1 previous $\oplus$ current		30.18	51.18

► [another slide with most-common pruning - still in training]

- ▶ Concatenation of sentences yield slight performance increases with regards to BLEU and TER metrics
 - ▷ preliminary results on WMT 2018 German→English ▶ [maybe most-common pruning yields better results than baseline - wait till training is done]
- ▶ Simple compression by pruning words according to frequencies

Future work:

- ▶ Implementation of multi-encoder approaches [Bawden & Sennrich⁺ 18, Voita & Sennrich⁺ 18]
- ▶ Test approaches in discourse-heavy data sets like OpenSubtitles [Lison & Tiedemann 16]
- ▶ Consider global context information

Thank you for your attention

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